Generalised Estimating Equations

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Introduction

Generalised estimating equations (GEE) fit marginal regression models for clustered data — repeated measures, longitudinal outcomes, familial correlations — without specifying random effects. Where mixed models target subject-specific (conditional) effects, GEE targets population-averaged (marginal) effects, answering the question "what is the average effect across the population?" rather than "what is the effect for a typical subject?" The two interpretations coincide for linear models with identity link but diverge for non-linear links (logistic, log).

Prerequisites

A working understanding of generalised linear models, the difference between conditional and marginal effects, and basic familiarity with cluster-robust standard errors.

Theory

GEE specifies a mean structure $g(\mu_{ij}) = X_{ij}^\top \beta$ for observation j in cluster i and a *working correlation* matrix $R(\alpha)$ governing within-cluster dependence. Standard choices:

- **Independence**: ignore correlation; coefficients still consistent.
- **Exchangeable**: equal correlation between all pairs within cluster.
- **AR(1)**: correlation decays geometrically with time gap.
- **Unstructured**: each pairwise correlation estimated separately.

Coefficient estimates are consistent even under correlation misspecification; standard errors use the sandwich (robust) variance estimator. The working-correlation choice affects efficiency but not asymptotic consistency.

QIC (quasi-likelihood information criterion) generalises AIC for GEE and is used for selecting both the working correlation and the mean structure.

Assumptions

Independent clusters, correctly specified mean model, and missing-completely-at-random (MCAR) for missing observations. The MCAR requirement is stricter than the MAR assumption of mixed models, so heavy missingness can favour the latter.

R Implementation

```r
library(geepack)

data(respiratory, package = "geepack")
fit <- geeglm(outcome ~ treat + age + sex, id = id, data = respiratory,
              family = binomial, corstr = "exchangeable")
summary(fit)

# AR(1) correlation
fit_ar <- geeglm(outcome ~ treat + age + sex, id = id, data = respiratory,
```

```r
                    family = binomial, corstr = "ar1")

# QIC for correlation structure selection
QIC(fit); QIC(fit_ar)
```

Output & Results

`summary(fit)` returns marginal coefficients with sandwich-robust standard errors and Wald z-tests. `QIC()` provides a quasi-likelihood information criterion for comparing models with different working correlations or different mean structures.

Interpretation

A reporting sentence: "A GEE logistic regression with exchangeable working correlation estimated the population-averaged odds ratio for treatment at 3.7 (95 % CI 2.0 to 6.6, $p < 0.001$); standard errors used the sandwich variance estimator to protect against working-correlation misspecification, and QIC favoured exchangeable over AR(1) ($\Delta\text{QIC} = 4$)." Always specify the working-correlation structure and use sandwich SEs.

Practical Tips

- Sandwich SEs are the standard inferential tool for GEE; do not rely on naive model-based SEs.
- Use QIC for selecting the working correlation; AIC and BIC do not apply because GEE is quasi-likelihood, not likelihood.
- Choose between mixed models and GEE based on the inferential question: mixed for subject-specific effects, GEE for population-averaged.
- For longitudinal binary outcomes, exchangeable working correlation is a sensible default; AR(1) when measurements are evenly spaced and correlation decays in time.
- For small numbers of clusters (< 40), sandwich SEs can be biased downward; use small-sample corrections (`geesmv` package) or fall back on mixed models with appropriate denominator df.
- Missing data: GEE requires MCAR for unbiased estimation, while mixed models tolerate MAR; consider this trade-off when choosing.

R Packages Used

`geepack` for `geeglm()` and QIC; `gee` for an alternative implementation; `geeM` for high-performance GEE on large datasets; `geesmv` for small-sample SE corrections; `multgee` for GEE with multinomial outcomes.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Correlation vs regression; Model-I/II regression
- Simple linear regression
- Multiple regression: confounding, interaction, centring
- Diagnostics: residuals, QQ, leverage, Cook's distance, VIF
- Robust regression, weighted LS, HC (sandwich) standard errors

- One-way ANOVA and contrasts (emmeans)
- Two-way / factorial ANOVA with interaction
- RCBD and repeated measures → mixed models
- GAMs with mgcv
- Non-linear regression with nls
- Logistic regression (binomial GLM)
- ANCOVA in RCTs (baseline adjustment)
- Ordinal and multinomial regression
- Poisson and negative-binomial regression
- Calibration, discrimination, ROC/AUC, Brier score
- Dichotomisation, change scores, regression to the mean
- Decision curves, NRI, IDI